

PH142: Introduction to Probability and Statistics in Biology and Public Health Spring 2026

Course format: Live online lectures, in-person labs and exams
4 semester credits

Course Description

This course is an introduction to statistics and data science, primarily for MPH and undergraduate public health majors, and others interested in public health topics.

The course can be divided into three parts.

In Part I, we will focus on learning to use R to explore datasets and summarize univariate and bivariate distributions visually and numerically. We will also introduce concepts around sampling and how study data are acquired. Specifically, we will use the dplyr and ggplot2 packages to manipulate and visualize data sets in R.

Part II of the course introduces classical problems in probability and the Normal, binomial, and Poisson distributions. We also introduce concepts around sampling variability and the Central Limit Theorem.

In Part III, we introduce statistical inference, the process of estimating statistics from samples to make inference about populations.

During all parts of the course we will use real and simulated data sets to gain experience conducting biostatistical analyses using R. We will follow the PPDAC model, which stands for "Problem, Plan, Data, Analysis, and Conclusion".

Prerequisites

No specific coursework is required, and no programming background is necessary. We expect a level of math equivalent to a US secondary school algebra course.

Course Learning Objectives

After successfully completing Part I of the course, you will be able to:

- Extract relevant statistical information from published articles in the scientific and popular press
- Describe distributions of variables visually and calculate summary statistics for measures of centrality and spread
- Determine the appropriate graphic to plot distributions and provide R code to manipulate and visualize data frames
- Identify basic sampling strategies and study designs used in Public Health
- Describe core concepts of ethics in Public Health
- Perform basic data manipulation in R

After successfully completing Part II of the course, you will be able to:

- Compute probabilities using the general rules
- Understand the concept of statistical independence
- Identify and describe binomial and Poisson random variables
- Compute probabilities using basic properties of the Normal distribution
- Express epidemiologic measures as probabilities
- Describe sampling variability and the central limit theorem
- Write R code to compute probabilities for the Normal, binomial, and Poisson distributions

After successfully completing Part III of the course, you will be able to:

- Estimate means, proportions, and differences between means and proportions, compute their confidence intervals and perform statistical tests
- Chose an appropriate measure and statistical test for the association between two variables
- State the assumptions and importance of the assumptions for statistical tests
- Describe and check the assumptions for simple linear regression.
- Interpret the confidence interval and statistical test of regression intercept and slope coefficients
- Demonstrate knowledge that has been used throughout the term, in terms of data visualization and data manipulation
- Write R code snippets to generate confidence intervals, perform hypothesis tests and calculate p-values
- Formulate a clear statistical question, address it with an appropriate method and accurately and clearly report your findings

Instructor Information and Communication

Course Instructor

Instructor **Mi-Suk Kang Dufour (she/her/hers)**

Email: mi-suk@berkeley.edu

Instructor Availability: <https://mi-suk.youcanbook.me/>

Graduate Student Instructors (GSIs)

While the instructor will interact with the whole class and will oversee all activities and grading, as well as being available to resolve any issues that may arise, the GSIs will be your main point of contact. Your GSIs are responsible for assisting you directly with questions about assignments and course requirements. They will lead lab sections and will host office hours. They will also facilitate ongoing discussion and interaction with you on major topics in each module. Communication with the teaching team regarding course content should be directed towards the ed discussion board.

Office Hours

Each GSI will hold office hours at least once per week. GSI office hours allow for synchronous interaction with the instructor and GSIs and are a good opportunity to discuss your questions relevant to the course. You may attend the office hours of any GSI regardless of your lab section. Scheduling of office hours will be determined according to student preferences following the beginning of term needs assessment.

The GSIs will offer office hours on campus. Some office hours may also be offered via Zoom. GSI office hours and locations will be posted to the course calendar. Instructor office hours will primarily be offered by appointment – please use the link <https://mi-suk.youcanbook.me/> to schedule a meeting with the instructor.

Course Mail

We strive to reduce email as much as possible. All questions about course material should be asked on Ed. This allows us to reduce email and also allows other students to benefit from the questions and answers. We will not answer any questions about course material via email. Email the GSI account to: make DSP accommodations for tests

or homework or request an assignment extension (see “Late Assignments” below). Email both the instructor and GSI account for personal concerns or disruptions that affect your performance in the course or during an emergency that will result or has resulted in a missed test. Do not use the bCourses emailing system.

Course Help

You’re not alone in this course; the instructor and GSIs are here to support you as you learn the material. It's expected that some aspects of this course will take time to grasp, and the best way to grasp challenging material is to ask questions.

Questions during lecture and lab are strongly encouraged. If something is unclear to you, it is probably unclear to many others in the room. There may be times, however, when the instructor or the GSI decides that a particular question or discussion is not helpful to the entire class or will take too long to address satisfactorily. In these cases, we may defer the question to be answered outside of class time. If important questions are answered outside of class, the answers will be shared with the entire class on Ed.

We will use Ed for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, the GSIs, and the instructor. In general, you can expect that the GSIs will respond to posted questions within 24 hours. Please do not email course content questions directly to the instructor or GSIs. The instructor and GSIs will not respond to questions about course content by email. **GSIs will not respond to Ed questions during holidays and breaks and you should expect longer turnaround times on the weekend (48 hours compared to 24 hours).** GSIs will respond to Ed questions up until 24 hours before exams. However, students may continue to post and answer each other’s questions during breaks and in the last 24 hours before an exam. For questions and concerns that are not related to course content, please email ph142@berkeley.edu. GSIs will do their best to respond to the course email account within 2 business days. Email responses may be slower over weekends and breaks.

Students with Disabilities

The mission of the Disabled Students' Program (DSP) is to ensure that all students with disabilities have equal access to educational opportunities at UC Berkeley. The DSP offers a wide range of services, accommodations, and auxiliary services for students with disabilities. These services are individually designed and based on the specific

needs of each student as identified by DSP's Specialists. We will accommodate disabled students' needs according to DSP documentation; please notify the DSP if you require such accommodation (DSP will then contact the instructor). **Note that this may take several weeks, so please initiate this process ASAP so that any accommodations can be implemented in time for the first midterm exam.** Steps to the application process: <https://dsp.berkeley.edu/students/new-students>. If you require DSP accommodations for a test, please email the GSI email account at ph142@berkeley.edu with your request and write "DSP accommodation" in the subject heading as soon as you know accommodations are required. If your accommodation allows for extension on take-home assignments, we ask that you discuss your request no later than 24 hours after the assignment is posted.

Course Materials and Technical Requirements

Course website

To access the course website, go to

<https://ph142-ucb.github.io/sp26/>

Here you will find links to required and optional readings, the syllabus, assignment descriptions and additional course resources. The course website will contain the most up to date schedule and assignment information.

Required Materials

We will be using R, a statistical programming language, and RStudio, an integrated development environment on Datahub, a cloud computing environment created at Berkeley. Use of R, RStudio, and Datahub is required for homework assignments and lab exercises and requires an internet connection and web browser. You will learn how to use R, RStudio, and Datahub during the first week of classes. You can access datahub from the links on the course website.

Optional Materials

The course textbook is "The practice of statistics in the life sciences" by Brigitte Baldi and David S. Moore. The 4th edition is the latest one, but previous editions are fine. The textbook covers material more relevant to Parts II and III of the course. Less of the material in Part I

is covered in the textbook. The textbook is not required. It is possible to complete the course using course materials and we note that the instructors differ in some places from the opinions presented in the text. Where the textbook and course materials differ the course materials will take precedence.

Additional Materials

In addition, here are some free online resources available as supplementary material. We link to these specific resources in the lecture slides when applicable:

Learning statistics with R: <https://learningstatisticswithr.com/lsr-0.6.pdf>

OpenIntro Statistics: <https://drive.google.com/file/d/0B-DHaDEbiOGkc1RycUtIcUtIeI/view>

A ModernDive into R and the Tidyverse: <https://moderndive.com/9-hypothesis-testing.html#ht-infer>

Statistical Thinking for the 21st Century: <https://statsthinking21.github.io/statsthinking21-core-site/ci-effect-size-power.html#statistical-power>

R for Data Science: <https://r4ds.had.co.nz/data-visualisation.html>

Technical Support

UC Berkeley provides technology support for graduate and undergraduate students that may prove valuable in this course, including free software (software.berkeley.edu), device lending (technology.berkeley.edu/step) and the student helpdesk (studenttech.berkeley.edu/techsupport).

If you're having technical difficulties please alert one of the GSIs immediately. However, understand that neither the GSIs, nor the instructor can assist you with technical problems other than those related to the course software systems R/Rstudio/Datahub. You must call or email tech support to resolve any technical issues.

To contact tech support, click on the "Help" button on the bottom left of the global navigation menu in bCourses. Be sure to document all interactions (save emails and transaction numbers).

Learning Activities and Assignments

You're expected to fully participate in all the course activities described below. All times listed are Pacific Time.

Lectures

Synchronous video lectures will be held M, W, F from 8:10 to 9:00 am. Lectures will be recorded and made available as soon as possible following each session. Although lecture attendance is not recorded, attendance it is strongly recommended. Participation activities during lecture will count towards your course grade. Please keep in mind that there may be some delay between live lecture and the posting of the lecture videos. Lectures will primarily cover material in the lecture slides and handouts. Code used during lecture will be shared in markdown format through the datahub, slides will be shared in PDF format prior to lecture. Lectures will introduce concepts, highlight examples of these concepts in the popular and scientific press, and walk through some examples and calculations. Students will be muted and their videos turned off by default since there are so many of you! I encourage you to ask questions using the chat or by unmuting and asking a question aloud. Feel free to turn on your camera when you ask a question orally. Please note that questions asked orally become part of the lecture/lab recording.

Lab/Discussion Sections

Lab/discussion sections will be held each week. These sections will be held in person for 2 hours each. Lab sessions are not recorded. Lab sections will go through hands on coding activities in R, review key concepts and include group activities and support for group data projects.

Quizzes (x% [percentage of total grade])

After completing the lectures and reading assignments for the week, take the [weekly quiz]. The quizzes will be [description of the quiz materials, platform (i.e., bCourses, Gradescope, etc.), question types, time limit, etc.]. Quizzes are due by [time am/pm Pacific Time] on [days] and have a [x-minute] time limit.

If you have a Letter of Accommodation at UC Berkeley, confirm with your GSI that it has been received and accommodations have been

made. Every time you start a quiz or exam, check to confirm you have the correct time accommodation. If not, notify your GSI.

Participation (10%)

Throughout the term you will be asked to provide feedback, practice with/test course systems and participate in course discussions. Participation activities will occur during lecture and lab and will not be announced. These may include zoom polls, questions or discussion activities. Participation activities will be graded for completion only. Participation assignments will be announced on Ed. These will be also marked for completion only. You may miss one participation assignment without penalty

Homework Assignments/problem sets (0%)

These will be distributed as R markdown files on datahub on Wednesdays (no assignments will be released the weeks of midterm exams). **Homework will not be submitted for marks** and you are encouraged to work on it in groups if that is how you learn best. **Completing the homework is good preparation for the midterms and final examination.** All solutions will be posted on datahub each Tuesday after the homework was made available.

Lab exercises (10%)

Lab exercises are intended to practice concepts from lecture in a practical programming environment. They are intended to be completed during the lab section with support and troubleshooting help from the GSIs. If you are unable to attend lab on a given week you may complete the lab independently. Lab exercises are generally released on Wednesdays prior to the first lab section, and are due by the following Saturday at noon. Lab exercises are submitted through the datahub to gradescope and are graded on correct completion. This means the grading is all or nothing. Assignments which are fully complete and pass all tests for code correctness will be counted 100%. Incomplete or incorrect assignments are counted as a 0%. Assignments that are complete and correct which are submitted after the due date but within 24 hours of the due date will be accepted for 50%. You may miss one lab assignment without penalty.

Weekly quizzes (10%)

Weekly quizzes will be available from 9 am on Fridays until noon on the following Saturday. Quizzes will be relatively short and meant to encourage you to keep on top of weekly content. Once opened, you

will have 1 hour to complete the quiz. Quizzes will cover material through the Wednesday of the current week. Your lowest quiz score will be dropped.

Midterms I and II and final exam (midterms 15% each, final 20%)

There are two midterms and one final exam. **If you have a conflict with any of the exam dates, please email the instructor before February 1st so that we can discuss possible accommodations.** Appropriate accommodations for the midterm will be made for those with disabilities (please refer to the "Disabilities" section) please contact the instructor as soon as possible if your accommodation for extended time may create a conflict with another scheduled class. Please note that only in extremely rare circumstances such as illness (with a doctor's note) will the midterm be given to individual students after the scheduled examination date. Exams will cover the material presented in lecture, supplemental videos, discussion, and lab sections, including R coding syntax, unless otherwise noted.

Midterms and final policies. The midterm and final will allow a "cheat sheet" of notes. You may bring one page of handwritten or printed notes with you to the midterms and two pages of handwritten or printed notes with you to the final. If your notes are printed the font should be a minimum of 10 point font. Your notes page will be turned in with the exam. If you want to have your notes to use later you should make a separate copy to keep that you do not bring to the exam. You **may not** use the internet to search for the answers or inform your answers. Using the internet is **strictly prohibited** and any evidence of this may result in a 0 on the exam. While you take the exam, you are prohibited from discussing the test with anyone other than the PH142 instructional team. Evidence of cheating may result in a 0 on the test or further disciplinary action. Exams will be given in person, on paper and scanned into gradescope by the teaching team. We will strive to return graded examinations within one week of the exam date.

Data skills demonstration group project (20%)

The purpose of the group project will be to use public health or biological data that you find or have access to and use it to demonstrate the statistical concepts that you've learned throughout the course. We will assign you to a project group based on your lab section and when possible your preferences for topics and meeting times. You will be asked about your preferences at the beginning of the term.

Grading

Your final course grade will be calculated as follows:

Table 1: Final Grade Percentages

Category	Percentage of Grade	Notes
Weekly quizzes	10%	Lowest grade dropped
Participation	10%	You may miss one activity without penalty
Lab completion	10%	You may miss one lab without penalty
Midterm Exam I	15%	
Midterm Exam II	15%	
Final Exam	20%	
Data project	20%	Submitted in 3 parts

Strategies for Successful Learning

While we know that not all strategies are equally useful for all students, based on past experience with students in this course we recommend:

- attending lab consistently
- asking questions in lecture/lab
- studying in a group and taking turns explaining concepts to each other
- reviewing questions and concepts from quizzes and exams for which you lost points
- use the practice exams to familiarize yourself with the types of questions you will see in exams and to practice the material

View the [Center for Teaching and Learning's page on \(meta\)cognitive strategies for student learning](#) to help you effectively study the material.

Take Care of Yourself

Do your best to maintain a healthy lifestyle this semester by eating well, exercising, getting enough sleep, and taking time to recharge your mental health. Taking time to care for yourself, and avoiding burnout, will help you achieve your academic, professional, and personal goals.

If you start to feel overwhelmed, be kind to yourself and reach out for support. Remember that seeking help is a courageous thing to do—for yourself and for those who care about you.

[Support Resources](#) include emotional, physical, safety, social, and other basic wellbeing resources for students. Academic resources can be found at the [Student Learning Center](#) and [English Language Resource](#) sites. Berkeley's Office of Emergency Management has resources to [prepare for emergencies](#).

We also ask that you look out for your fellow peers. If you see any of the signs below that may indicate your classmate may need assistance, please use the resources above or reach out to any of the GSIs or Professors.

Withdrawing from other people

Changes in weight or eating patterns

Changes in sleeping patterns

Fatigue or lack of energy

Increased anxiety or irritability

Feeling worthless or hopeless

Course Policies

Due Dates

Due dates for all graded assignments will be listed on the course website and will be communicated weekly in announcements on the Ed discussion board.

Scheduling Conflicts

If you have a conflict with any of the midterm/exam dates, please email the instructor by February 1st so that we can discuss possible accommodations. I will try my best to help you with making accommodations but cannot promise them in all cases.

Grace Period

Due to the nature of electronic submission, we understand that some students may experience technical difficulties with submission close to the deadline for assignments that are uploaded. Therefore, we are offering a grace period of two hours, to account for these submission issues. If you are having issues during this time, please email your submission to the GSI email at (ph142@berkeley.edu) and are not able to submit within the grace period. **We will not be accepting requests regarding submission errors past the two hour grace period on the due date.** The grace period applies by default, you do not need to notify us to use it. Please also note that it is the student's responsibility to ensure that an assignment has been properly uploaded. We recommend that you open the gradescope assignment and confirm that your assignment has been submitted.

Late Work Policy

Assignments submitted after the grace period but within 24 hours after the due date will be accepted but will be penalized by 50%. Extensions can be made for DSP students but should be requested ideally before the due date by emailing the GSI email account. Anyone else requesting an exemption should email the GSI account explaining their situation. If an emergency event prevents submitting an assignment by the deadline, please contact the GSI email account as soon as reasonably possible, including documentation with your request for extension

Regrade policy

Regrades will be allowed on quizzes and midterm exams, and must be submitted **within three days** after the grades are released using Gradescope. Regrades are for the purposes of correcting an error in the application of the published rubric for the assignment. Regrades should only be submitted when the answer on the assignment has not been graded according to the rubric. If you have a question about the material and/or why the rubric answer is correct please speak to a member of the teaching team rather than submitting a regrade request. Note that if you request reconsideration of a graded question, instructors may reconsider grades on the entire assignment. Due to the short turnaround time for final grade submissions we generally cannot accommodate re-grade requests for the final exam.

Academic Integrity

You're a member of an academic community at one of the world's leading research universities. Berkeley creates knowledge that has a lasting impact in the world of ideas and on the lives of others; such knowledge can come from an undergraduate paper as well as the lab of an internationally known professor. One of the most important values of an academic community is the balance between the free flow of ideas and the respect for the intellectual property of others. Scholars and students always use proper citations in papers; professors may not circulate or publish student papers without the writer's permission; and students may not circulate or post materials (handouts, exams, syllabi—any class materials) from their classes without the written permission of the instructor.

Any test, paper or report submitted by you and that bears your name is presumed to be your own original work that has not previously been submitted for credit in another course unless you obtain prior written approval to do so from your instructor. In all of your assignments, including your homework or drafts of papers, you may use words or ideas written by other individuals in publications, websites, or other sources, but only with proper attribution. If you're unclear about the expectations for completing an assignment or taking a test or examination, be sure to seek clarification from your instructor or GSI beforehand. For additional information, read this UC Berkeley Library guide on [How to Avoid Plagiarism](#).

As a member of the campus community, you're expected to demonstrate integrity in all of your academic endeavors and will be evaluated on your own merits. The consequences of cheating and academic dishonesty—including a formal discipline file, possible loss of future internship, scholarship, or employment opportunities, and denial

of admission to graduate school—are simply not worth it. Read more about [Berkeley's Honor Code](#).

Incomplete Course Grade

Students who have substantially completed the course but for serious extenuating circumstances, are unable to complete the remaining course activities, may request an Incomplete grade. This request must be submitted in writing to the GSI and instructor. You must provide verifiable documentation for the seriousness of the extenuating circumstances.

Refer to the Office of the Registrar's website for more information on the university's policy on [Incomplete Grades](#).

End of Course Evaluation

UC Berkeley is committed to improving our online courses and instruction. Before your course ends, please take a few minutes to participate in the course evaluation. We are interested in your online learning experience, and your feedback will help us plan for the future and make improvements. The evaluation does not request any personal information, and your responses will remain strictly confidential.

Course Schedule